The Chain of Karma

The icosahedron and the dodecahedron as endless shells of each other

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Version 2 · 12 October 2026 · DOI 10.5281/zenodo.23310542 · Licence CC BY 4.0 · Data, scripts, renders and checks are included with this record and at https://github.com/DICTOR-Master/kaleidohedra.
The Lotus seed, the Chain of Karma, the FUJI tip + mirror rule and the names Lotus seed, AXE, FUJI and CLEO are DICTO's. Design study and checking with Claude (Anthropic).

Abstract. We show that the regular icosahedron and the regular dodecahedron can each be filled exactly as a shell around the other, and that the two shells alternate without end. A dodecahedron of edge 1 is an icosahedron of edge 1/φ inside a shell of 12 pentagonal pyramids, 30 AXE and 20 CLEO; an icosahedron of edge φ² (the Lotus seed) is that dodecahedron inside a shell of 12 J11 (each an icosahedron with one pyramid cap removed), 30 AXE and 20 FUJI. Thus the nesting I(1/φ) ⊂ D(1) ⊂ I(φ²) ⊂ D(φ³) ⊂ … grows by φ³ per cycle, and each shell has one piece for every vertex, edge and face of the dual pair (12, 30, 20). The pieces are exact, at edge 1: AXE is a tetrahedron with five edges 1 and one 1/φ (two equilateral faces, two golden triangles 1, 1, 1/φ); CLEO is a triangular pyramid on an equilateral base of side 1/φ with side edges 1; FUJI is a pyramid of height 1/√3 on a flat-ended (truncated) triangle with sides 1 and 1/φ alternating and all angles 120°, with side edges 1, three equilateral and three golden faces, and a cube corner at its apex; the pentagonal pyramid has all edges 1. Every dihedral angle of the kit is, modulo 180°, an integer combination of two angles, α = arccos(√5/3) = 41.81° and δ = arctan(3 − √5) = 37.38°: the icosahedron's own is 180° − α = 138.19°, and FUJI's faces meet its base at α on the long sides and δ on the short. Because every icosahedron inside is again whole, each can be replaced by its own Lotus seed: the Chain of Karma, built and checked to 20,497 pieces. Only the icosahedron is ever subdivided: the gap pieces stay whole (they cannot be cut into the kit at one size, by the Dehn invariant) and occur at every scale φᵏ. A second exact cycle, through the icosidodecahedron, the fullerene cage and the rhombic triacontahedron, shares the dodecahedron with the Lotus seed and is described separately (Spheres of Dharma). The nesting of the dual pair at golden ratios goes back to Pacioli and Kepler and appears at every φ-power in Gray (2002); a different dodecahedral growth, with whole dodecahedra inside, is due to Al Raisi, Koca, Koca and Koc (2026). The exact shells and their endless alternation are new.

1. Two shells

Throughout, φ = (1 + √5)/2, edges are given relative to the dodecahedron D(1) of edge 1, and I(a), D(a) denote the regular icosahedron and dodecahedron of edge a. Two exact decompositions carry the whole construction:

$$D(1) = I(1/\varphi) + 12\ J2 + 30\ AXE + 20\ CLEO, \quad I(\varphi^2) = D(1) + 12\ J11 + 30\ AXE + 20\ FUJI.$$

In the first, the 12 pentagonal pyramids J2 (all edges 1) stand on the dodecahedron's faces with their points inward, and each point lands exactly on a corner of I(1/φ): the dodecahedron's inradius 1.113516 minus the pyramid's height 0.525731 is 0.587785, the circumradius of I(1/φ). In the second, each of the 12 outer icosahedra I(1) is cut by a face plane of D(1) into a J2 cap (inside D) and a J11 (outside). Volumes: 0.515028 + 12 × 0.301503

+ 30 × 0.083333 + 20 × 0.051503 = 7.663119 = D(1), and 7.663119 + 12 × 1.880192 + 30 × 0.083333 + 20 × 0.321175 = 39.148928 = I(φ^2) = φ^6 I(1). Together they give the Lotus seed (DICTO Icosa-13, 113 pieces): I(φ^2) = 12 I(1) + I(1/φ) + 60 AXE + 20 FUJI + 20 CLEO, whose 13 icosahedra are arranged as in Gray (2002). Each shell has one piece for every vertex, edge and face of the dual pair: 12, 30 and 20.

2. The Chain of Karma

Since the outer form of the Lotus seed is exactly I(φ^2), it is a golden dissection rule for the icosahedron. Replacing every icosahedron by a Lotus seed of its own size and turn, and keeping AXE, FUJI and CLEO whole, fills I(φ^4), I(φ^6), ... exactly, upward without end, and in the same way downward. After n steps there are 13^n icosahedra and $100(13^n - 1)/12$ gap pieces, AXE, FUJI and CLEO in the ratio 3 : 1 : 1. Steps 1 to 3 were built and checked: 113, 1,569 and 20,497 pieces, volume equal to φ^{6n} times I(1) to within 3×10^{-9}. The icosahedra's share of the volume falls by $(12 + \varphi^{-3})/\varphi^6 = 0.6823$ each step (68.19%, 46.50%, 31.71%). Keeping only the icosahedra gives a fractal of similarity dimension D with $12\ \varphi^{-2D} + \varphi^{-3D} = 1$, D ≈ 2.6063, slightly above the Sierpinski icosahedron's log 12 / log φ^2 ≈ 2.5819 (12 corner copies only).

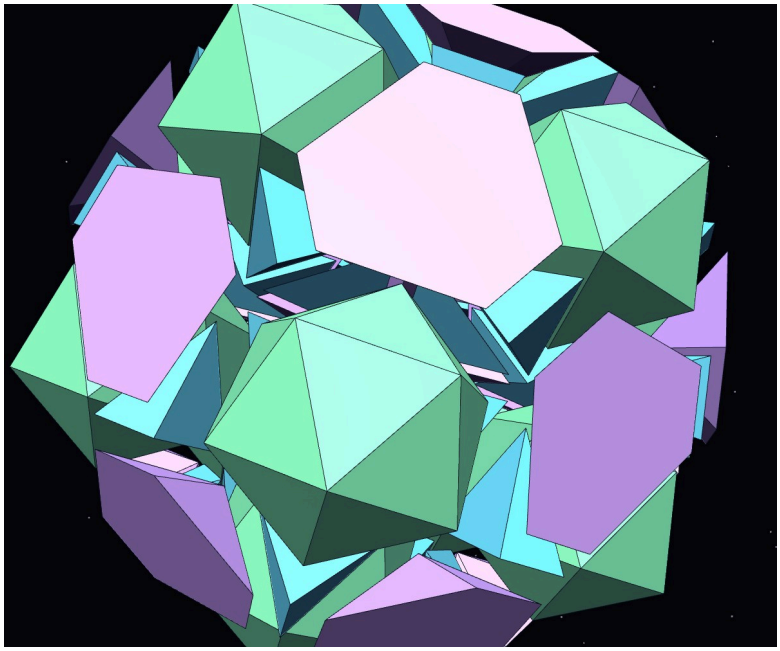


Figure 1. The Lotus seed pulled apart.

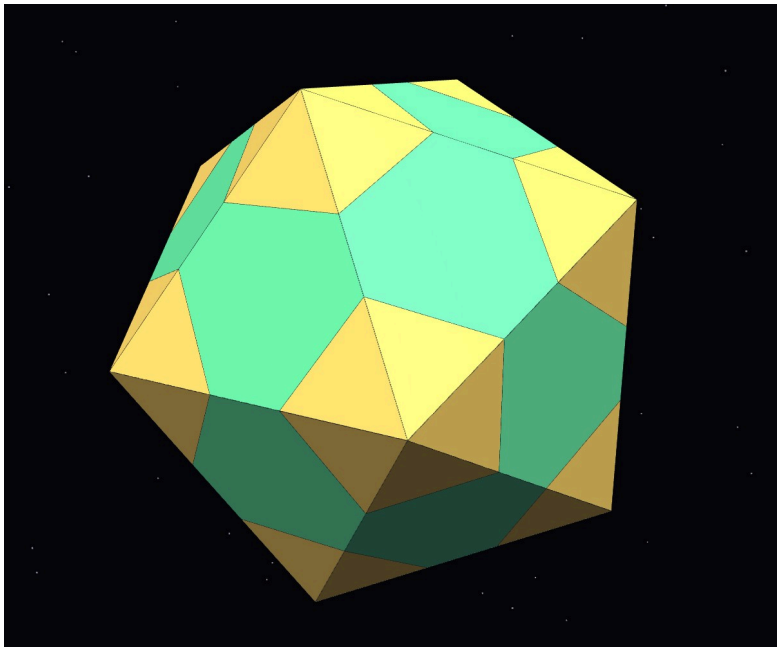


Figure 2. The Lotus seed closed: its surface carries the fullerene (C60) pattern, with the 20 flat-ended triangles (sides 1 and 1/φ alternating, all angles 120°) in green.

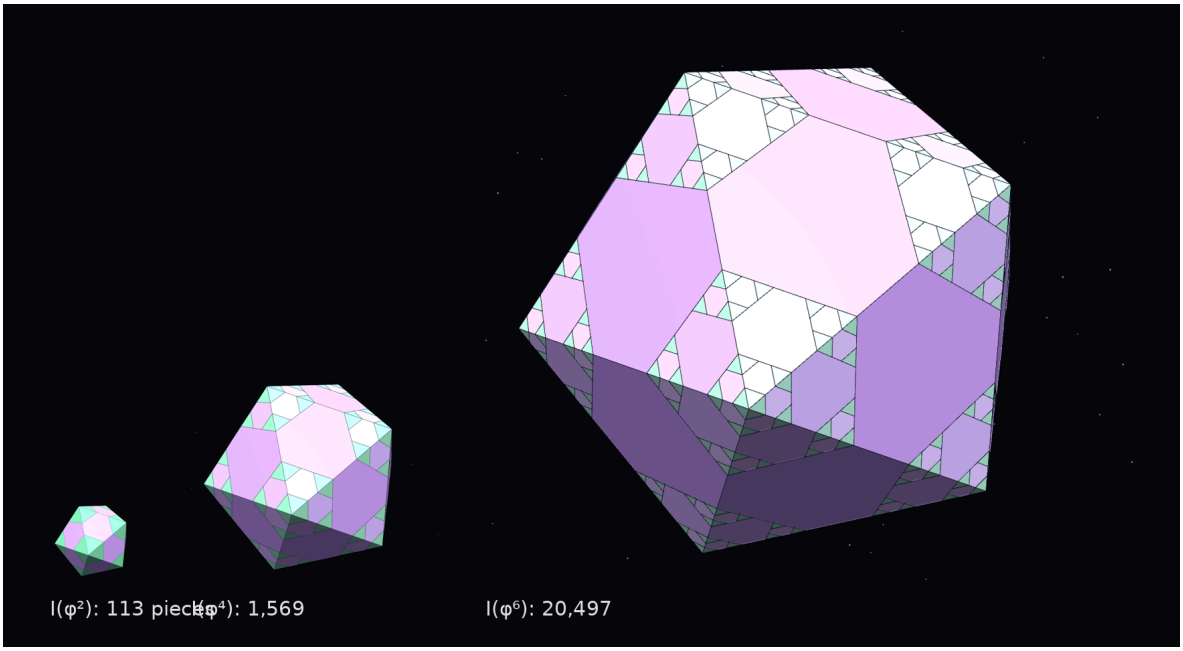


Figure 3. The Chain of Karma growing, at true relative size: the Lotus seed I(φ²), step 2 I(φ⁴) and step 3 I(φ⁶). Each step replaces every icosahedron by a Lotus seed of its own size; the pieces keep their sizes and the whole grows by φ² per step.

3. The gap pieces and the Dehn invariant

Every dihedral angle of the kit is, modulo 180°, an integer combination of α = arccos(√5/3) = 41.810° and δ = arctan(3 − √5) = 37.377°. This makes the Dehn invariant computable exactly (coordinates in Q(√5), integer relations by PSLQ). At a single size the gap pieces cannot be rebuilt from the kit: φ²·AXE, φ²·FUJI and φ²·CLEO pass the volume test (for example φ²·AXE has the volume of 9 AXE + 2 FUJI + 2 CLEO) but no whole-number combination of the edge-1 kit matches their Dehn invariants; the same program recovers the Lotus seed's own counts for φ²·I as a check. So the Chain of Karma is a hierarchy with gap pieces in every size step of φ², not a single-size tiling. With mixed sizes the screen rules nothing out; it is a necessary

condition only. AXE and CLEO are, up to the factor 1/φ, the Mosseri–Sadoc tetrahedra t6 and t4 (Mosseri and Sadoc 1982; Kramer 1982, whose seven tiles add T2 = t2 ∪ t4). The six tetrahedra admit no stone inflation (Papadopolos and Ogievetsky 1999).

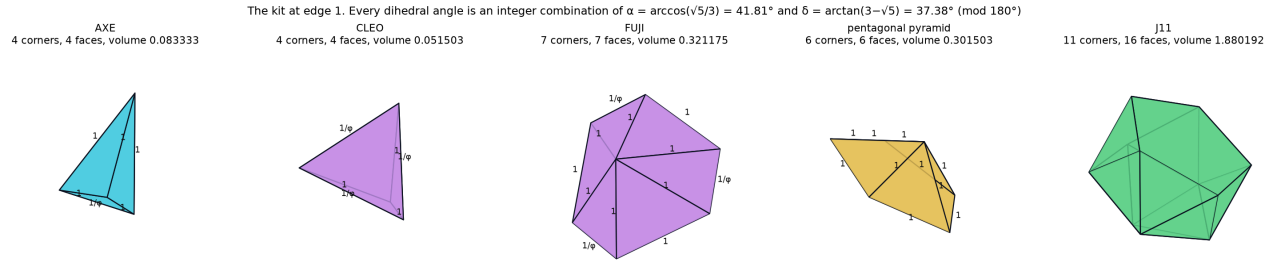


Figure 4. The kit at edge 1, every edge labelled: AXE, CLEO, FUJI, the pentagonal pyramid and J11, with their volumes. 60 AXE + 20 FUJI + 20 CLEO + 12 icosahedra of edge 1 + 1 of edge 1/φ fill the icosahedron of edge φ² exactly.

4. FUJI

FUJI is a pyramid of height 1/√3 on a flat-ended triangle (sides 1 and 1/φ alternating, all angles 120°) with side edges 1; its apex is a corner of the cube inside the dodecahedron. It splits as that cube corner (edges 1, 1, 1 at right angles; volume 1/6) plus three pieces each of CLEO's volume (1/6 + 3 × 0.051503 = 0.321175), and into 12 right-angled pieces of two kinds whose faces slope at exactly 41.81° and 37.38°, the kit's two angles.

5. The dodecahedral Chain

DICTO's dodecahedral Chain uses the first decomposition, D(1) = 12 J2 + I(1/φ) + 30 AXE + 20 CLEO, at every scale, with its central icosahedron replaced by the Chain of Karma: $D(\varphi^{2n})$ is the Chain-filled $I(\varphi^{2n-1})$ with 12 J2, 30 AXE and 20 CLEO of the stage's size around it, 63, 175 and 1,631 pieces for D(1), $D(\varphi^2)$ and $D(\varphi^4)$. Al Raisi, Koca, Koca and Koc (2026) grow the dodecahedron with whole dodecahedra inside; here it grows around icosahedra.

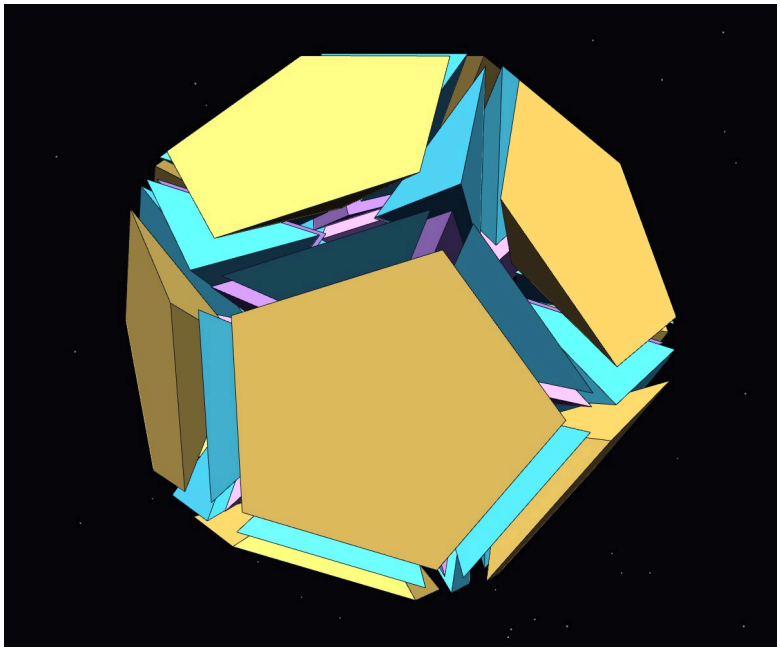


Figure 5. The dodecahedral Chain, its shell pulled apart.

6. A second cycle: the Spheres of Dharma

A second exact, endlessly repeating cycle passes through the icosidodecahedron, the fullerene cage and the rhombic triacontahedron: icosahedron φ^2 ⊃ triacontahedron ⊃ cage ⊃ dodecahedron ⊃ icosidodecahedron ⊃ icosahedron $1/\varphi$. It shares the dodecahedron of edge 1 and both icosahedra with the Lotus seed, and each half of either structure fills its side of that dodecahedron exactly, so the two can be exchanged half by half at every scale. It is described in its own paper (DICTO, Spheres of Dharma, #23).

7. The 600-cell, flattened

DICTO's route into four dimensions began with the DICTO Dodeca-13 (#13): a dodecahedron with twelve more on its faces, the 120-cell's first shell flattened, its 10.3° gaps being the flattening. The Lotus seed is the same for the 600-cell, the 120-cell's dual, as the icosahedron is the dodecahedron's. Seen down one vertex, the 600-cell (circumradius 1, edge $1/\varphi$) has shells of 1, 12 (icosahedron, radius 0.587785), 20 (dodecahedron), 12 (icosahedron), 30 (icosidodecahedron), …, alternating icosahedron and dodecahedron as the Chain's shells do. Two matches are exact (`cell600_match.py`): the Lotus seed's small icosahedron is the 600-cell's first shell (edge $1/\varphi$, circumradius 0.587785), and the centres of the Lotus seed's 13 icosahedra are the 600-cell's centre vertex and its 12 neighbours, scaled by φ^2 (radius 1.538842, spacing φ). The 600-cell is the icosahedral "12 around 1" packing that closes only in curved four-dimensional space; in flat space twelve around one leaves gaps (geometrical frustration, Sadoc and Mosseri). The Lotus seed reads as a flattened piece of the 600-cell, with AXE, FUJI and CLEO as exactly what flat space needs where curved space closes by itself, and the Chain of Karma repeats that flattening at every scale. The match is in projection; we do not claim the Lotus seed is a projection of the 600-cell.

The fullerene chain

DICTO (2026-10-11): the Lotus seed shows a fullerene architecture. Checked (`fullerene_surface.py`): its surface is 60 triangles of edge 1, grouped round the 12 corners as pentagonal caps, and 20 flat-ended triangles (DICTO's term: triangles with their corners cut flat, sides 1 and $1/\varphi$, the FUJI bases) in the places of the hexagons, so the Lotus seed has the face pattern of C60, the buckyball (Kroto et al. 1985), with flat-ended triangles for its hexagons: a truncated icosahedron whose cut corners are filled back by whole icosahedra. Chain step 2 keeps the 12 caps, has flat-ended triangles in two sizes (60 small, 20 of sides φ and φ^2) and 60 pairs of triangles folded over the edges: fullerenes inside fullerenes, every piece a triangle, whole or flat-ended. Counted as hexagons it has C180's 80, but it is not a Goldberg polyhedron (Goldberg 1937). The nesting recalls carbon onions, nested icosahedral fullerene shells (Ugarte 1992).

8. Periodic and aperiodic

The Chain of Karma is self-similar and not periodic: its symmetry is icosahedral and its scale steps are powers of φ. A periodic frame for icosahedra exists: an icosahedron with six edge roofs is a regular octahedron, so icosahedra with their roofs in the octahedra of the octet truss, and smaller icosahedra with corner and edge caps in its tetrahedra, fill space exactly (DICTO's icosahedral octet). Whether the Chain can grow inside that frame, joining the periodic and the self-similar orders, is open.

9. Self-similar, repeating, not periodic

Scaling is the growth step: the icosahedron and dodecahedron shells repeat by scaling, φ^3 per cycle, and the counts follow it exactly (13^n icosahedra after n steps of the Chain, gap pieces in the fixed proportion AXE : FUJI : CLEO = 3 : 1 : 1), as the growth matrix's largest (Perron–Frobenius) eigenvalue and its eigenvector say. **Fit** is the exact fit: each shell fills the space round the other, with volumes and Dehn invariants balancing. **Non-periodicity** is that no sliding of the whole lands it on itself. The three are linked: because the scale factor is irrational, a sliding repeat would force repeats at φ^3, φ^6, … times every period, which no single lattice of periods allows; and the same scaling makes every local arrangement recur.

Status. Proved: the shells, the exact fits and the counts. The Chain as built has a centre, so it is trivially not periodic. It is an iterated (hierarchical) dissection in which only the icosahedron is subdivided, not a substitution tiling in the strict sense, which would require every prototile to be subdivided; pieces of every size occur.

10. Relation to known work, and status

Nesting of the icosahedron and dodecahedron. Golden nestings of the two go back to Pacioli and Kepler. Gray (2002) describes icosahedra at every φ-power, each containing a central icosahedron at φ^{-3} and twelve at φ^{-2} (the arrangement of the Lotus seed's icosahedra), and dodecahedra that "scale pack" with them; the gaps are not filled. The Sierpinski icosahedron keeps the twelve corner icosahedra at $1/\varphi^2$ and leaves the gaps empty. Al Raisi, Koca, Koca and Koc (2026) grow the dodecahedron by τ^n with whole dodecahedra inside, using modified Mosseri–Sadoc tiles, and note that icosahedra cannot be treated this way with those tiles.

Tiles. AXE and CLEO are the Mosseri–Sadoc tetrahedra t6 and t4 at scale $1/\varphi$ (Mosseri and Sadoc 1982; Kramer 1982); the six tetrahedra admit no stone inflation (Papadopolos and Ogievetsky 1999; Ogievetsky and Papadopolos 1999), and Koca et al. (2020) state that they do not inflate the icosahedron past edge τ.

Nested icosahedral shells in matter. Concentric icosahedral shells are central to the structure of real materials and viruses, as point arrays: Mackay's icosahedral sphere packing (1962) and Frank's icosahedral order in liquids (1952); the Tsai-type cluster of icosahedral quasicrystals and their approximants, concentric dodecahedron, icosahedron and icosidodecahedron shells inside a rhombic triacontahedral envelope (Takakura et al. 2007; Deguchi et al. 2015; Ishimasa 2019); and icosahedral viruses (Caspar and Klug 1962), whose capsids and genomes follow nested icosahedral point arrays at fixed radial ratios, obtained from affine extensions of the icosahedral group or by projection from a six-dimensional lattice (Keef and Twarock 2009; Twarock, Valiunas and Zappa 2015), the same framework applying to nested fullerene cages (Dechant et al. 2014). The Chain of Karma differs in kind: its shells are solid and filled exactly, with whole regular icosahedra among the pieces, and the icosahedron and dodecahedron alternate as shells of each other without end.

Status. In the sources above and those in the study folder we found no description of this solid, exact, alternating filling. The search is not exhaustive; the claim of novelty is limited to it.

Remarks

The following are DICTO's own readings of the work, given as remarks; they are not mathematical claims.

On the two shells: "It is like yin and yang: neither one nor the other." Each solid holds the other at its centre, and each is the other's shell.

On the 600-cell: "It does strike me as remarkably similar to a 4D polytope."

On section 9: "The pattern that doesn't repeat is exactly the pattern that is repeating." DICTO relates the Chain to Kyushindo, the philosophy of Kenshiro Abbe, with its three principles 万物流転 *banbutsu ruten* (all things in constant flux), 律動 *ritsudō* (rhythmic motion) and 調和 *chōwa* (harmony): change, rhythm and harmony, each contained in the others. In section 9's terms they correspond to non-periodicity, scaling and exact fit.

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